

SENDAI, Japan

WMO: 475900

Lat: 38.2619N

Lon: 140.8967E

Elev: 144

StdP: 14.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.6	27.4	12.7	10.6	34.8	14.7	11.6	35.1	27.8	37.2	24.8	35.6	6.8	340	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	89.0	76.2	86.3	75.3	83.8	74.2	77.9	85.0	76.9	83.3	75.9	81.7	8.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.0	136.7	81.2	75.0	132.1	80.6	74.1	127.7	79.8	41.5	85.3	40.5	83.5	39.6	82.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.2	18.8	16.2		DB	22.7	93.9	2.2	2.2	21.1	95.5	19.8	96.8	18.5	98.0	16.9	99.6
				WB	20.6	80.1	1.8	1.3	19.3	81.0	18.2	81.8	17.3	82.5	16.0	83.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	35.5	36.5	41.9	51.5	60.6	66.9	74.3	76.8	70.6	60.6	49.6	40.3	(6)
	DBStd	15.19	4.16	4.63	5.48	5.97	5.39	4.86	5.66	5.13	5.39	5.19	5.69	5.02	(7)
	HDD50	1520	448	378	258	52	2	0	0	0	0	1	78	303	(8)
	HDD65	4418	913	798	715	406	156	33	3	0	11	154	463	766	(9)
	CDD50	3541	0	0	8	97	329	508	752	831	618	331	65	2	(10)
	CDD65	965	0	0	0	1	18	91	290	366	180	19	0	0	(11)
	CDH74	5864	0	0	0	6	85	307	1888	2717	820	41	0	0	(12)
	CDH80	1573	0	0	0	0	14	47	544	816	150	2	0	0	(13)
(14) Wind	WSAvg	7.1	7.9	8.2	8.5	8.1	7.1	6.2	5.5	5.7	6.3	6.8	7.1	7.6	(14)
(15) Precipitation	PrecAvg	48.9	1.7	1.7	2.8	3.6	4.4	5.5	6.3	6.4	7.5	4.8	2.6	1.5	(15)
	PrecMax	70.8	5.8	5.5	7.3	8.5	8.8	11.1	17.5	17.5	25.2	13.7	9.6	6.6	(16)
	PrecMin	32.1	0.1	0.1	0.1	0.2	1.1	2.0	1.1	0.7	0.5	0.3	0.2	0.1	(17)
	PrecStd	8.4	1.3	1.2	1.6	1.9	1.8	2.1	3.5	4.2	4.5	3.1	1.9	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	55.7	64.1	74.0	81.8	85.1	91.3	93.2	88.0	78.6	67.3	58.6	(19)
		MCWB	44.7	45.5	49.9	56.4	61.9	69.7	76.2	77.0	74.9	66.6	56.3	50.9	(20)
	2%	DB	47.6	50.8	58.9	68.6	75.8	80.5	88.4	90.0	84.1	74.2	64.2	54.7	(21)
		MCWB	40.2	42.8	47.8	54.2	60.7	68.1	75.7	76.9	73.1	64.5	55.1	47.6	(22)
	5%	DB	44.7	47.2	54.8	65.1	72.5	77.3	85.7	87.2	81.0	71.3	61.5	51.2	(23)
		MCWB	38.2	40.1	45.7	52.9	59.7	66.9	75.3	76.3	71.8	62.7	53.9	44.7	(24)
	10%	DB	42.5	44.2	51.4	61.6	69.5	74.7	83.0	84.7	78.6	68.9	59.0	48.5	(25)
		MCWB	36.5	37.9	43.6	51.7	58.7	66.3	74.0	75.6	71.5	60.9	52.6	42.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	47.6	52.9	62.0	66.8	74.1	78.6	79.9	77.3	70.4	60.1	52.6	(27)
		MCDB	50.4	53.2	59.2	66.6	73.8	79.8	86.6	88.4	82.3	74.7	63.2	56.6	(28)
	2%	WB	41.4	43.7	49.8	57.6	64.6	71.2	77.3	78.5	75.9	67.1	57.9	48.6	(29)
		MCDB	45.9	49.2	56.0	64.1	71.0	76.7	84.9	85.8	80.6	71.1	61.5	53.2	(30)
	5%	WB	39.2	41.1	47.4	55.5	62.6	69.4	76.3	77.5	74.6	64.7	55.5	45.6	(31)
		MCDB	43.8	46.3	53.2	61.6	68.5	74.4	83.2	84.0	79.1	68.8	59.9	50.9	(32)
	10%	WB	37.0	38.6	44.6	53.4	61.0	67.7	75.2	76.6	73.0	62.7	53.2	42.9	(33)
		MCDB	41.8	43.6	50.8	59.1	66.4	72.5	81.3	82.5	77.3	67.4	57.9	47.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	12.1	13.9	14.8	13.6	11.0	10.3	10.6	11.1	12.9	13.5	11.8	(35)
		MCDBR	14.5	16.5	19.3	20.5	18.8	15.7	14.6	14.4	13.7	14.9	16.8	15.7	(36)
	5% WB	MCWBR	9.9	11.2	12.4	11.1	9.3	7.2	5.9	5.8	6.6	8.6	10.9	11.0	(37)
		MCWBR	14.4	16.3	17.9	16.9	15.1	12.4	13.0	12.7	11.3	12.0	15.1	15.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.375	0.420	0.494	0.544	0.564	0.562	0.571	0.558	0.487	0.443	0.406	0.382	(40)
		taud	2.090	1.970	1.862	1.819	1.827	1.907	1.947	1.974	2.134	2.158	2.168	2.140	(41)
	Ebn at Noon		243	247	240	238	236	233	232	241	239	232	230	230	(42)
		Edh at Noon	37	48	59	66	66	61	58	56	45	40	35	33	(43)
(44) All-Sky Solar Radiation	RadAvg		629	872	1169	1460	1600	1539	1395	1396	1152	930	677	534	(44)
	RadStd		42	59	82	142	160	97	213	174	98	91	45	44	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page